Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/17/2009
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Transgenic Plant Biomonitors of Space Flight Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	robferl@ufl.edu	**Fax:**	FY 352-392-4072
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1928
Organization Name:	University of Florida		
PI Address 1:	Horticultural Sciences, 1301 Fifield Hall		
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PI Web Page:	http://www.hos.ufl.edu/		
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/15/2007	**End Date:**	05/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-129 ; ISS NOTE: Extended to 5/31/2011 per NSSC info; original end date 3/14/2010 (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Co-PI: Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX07AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is designed to be the next-generation of our previous flight project, PGIM-01, and investigates the use of reporter gene technology for the molecular analysis of development and stress in plants within space flight environments. We have designed a series of transgenic plants that carry sensor promoter - reporter gene constructs that are capable of monitoring a variety of environmental and developmental influences that might be altered in microgravity and general space flight environments. The first generation of these experiments with the GUS reporter have been tested for biological veracity on KC-135 parabolic flight missions, and flew as PGIM-01 in the Plant Growth Facility on STS-93. Results of these KC-135 missions and STS-93 were used in this proposal to (1) design a second generation of sensor promoters to refine our definition of the cellular biological impact of microgravity and space flight; (2) design a second generation of reporter gene constructs that include real-time and non- destructive observations of reporter gene activity; (3) design and conduct short and longer term flight experiments that begin a comprehensive analysis of molecular genetic changes induced by space flight; and (4) develop methods to integrate reporter gene activities with data from DNA chip analysis of genome expression. The long-term goal is an understanding of the space flight induced molecular changes in plant gene activity, an understanding that will lead to hardware and/or genetic engineering modifications that will mitigate the negative impacts of space flight on plant growth and development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of signal transduction in higher plants (and by inference, all eukaryotes) is central to basic molecular biology research.
Task Progress:	The past year saw the further testing and development of the ABRS hardware and aspects of the TAGES biology that will populate the unit. Preparations for the Payload Verification Test (PVT), which is scheduled for December - January 2009, was a major focus for the year. The Imaging Hardware underwent significant improvements throughout 2008, however there are still issues (predominantly software) being resolved at the time of this report. This progress report will describe the continued work our laboratory conducted with the ABRS ground unit and LED systems at the University of Florida, as well as provide an evaluation of aspects of the GIS (GFP Imaging System) hardware and software. A C9 parabolic flight provided information signal transduction of early detection of changes in the gravity vector.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Paul A-L, Bamsey M, Berinstain A, Braham S, Neron P, Murdoch T, Graham T, Ferl RJ. "Deployment of a prototype plant GFP imager at the Arthur Clarke Mars Greenhouse of the Haughton Mars Project." Sensors 2008 Apr;8(4):2762-73. http://dx.doi.org/10.3390/s8042762 , Apr-2008
Articles in Peer-reviewed Journals	Paul A-L, Folta KM, Ferl RJ. "14-3-3 Proteins, red light, and photoperiodic flowering: A point of connection?" Plant Signaling and Behavior 2008 Aug;3(8):511-5. http://www.landesbioscience.com/journals/psb/article/PaulPSB3-8.pdf , Aug-2008
Articles in Peer-reviewed Journals	Folta KM, Paul A-L, Mayfield JD, Ferl RJ. "14-3-3 isoforms participate in red light signaling and photoperiodic flowering." Plant Signaling Behavior 2008 May;3(5):304-6. http://www.landesbioscience.com/journals/10/article/5288/ , May-2008